

Using an Interactive Bash Shell on the Analysis Cluster

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Login Node (submit0) Access

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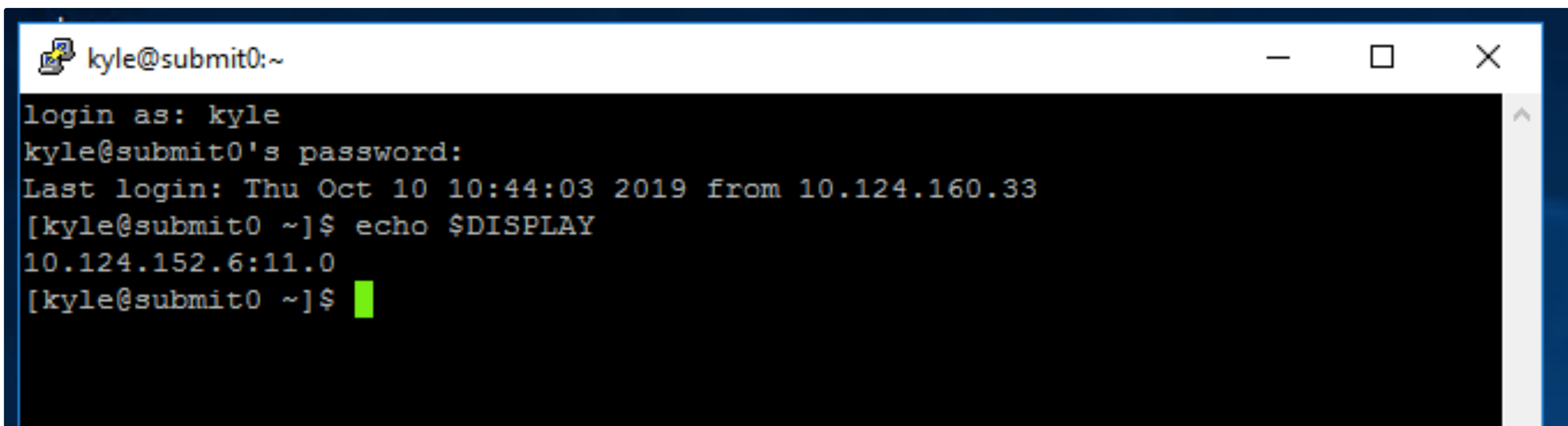
Windows

Pre-Requisites

- Install Putty and Xming [through Software Center](#)
- [Configure and save](#) a Putty session

Instructions

1. Start Xming if you don't already have it running
2. Open Putty and launch the saved session to submit0
3. Verify that your X11 forwarding is enabled by printing the DISPLAY variable (this will be empty if it isn't enabled)



4. Use **srund** to open an interactive bash shell with X11 forwarding on a compute node

```
1 srun --pty --x11 bash
2 # --pty opens an interact-able bash shell and --x11 enables X11 forwarding
```

```
submit0.c2.laureateinstitute.org - PuTTY
[aaron@submit0 ~]$ srun --pty --x11 bash
srun: error: x11: job has no allocated nodes defined
[aaron@compute207 ~]$
```

You will notice that the computer name changes once your interactive session begins.

5. From here you can load the AFNI module and then run the afni command to open the GUI, or run any other program you would like.

```
1 module load afni
2 afni
```

6. Close the AFNI window when you are done and enter the exit command into the command line when you are finished to exit your `srn` session.

Example

Example output

```
1 [user@submitt0 ~]$ srun --pty --x11 bash
2 srun: error: x11: job has no allocated nodes defined
3 [user@compute200 ~]$ module load afni
4 [user@compute200 ~]$ afni
5 ...
6 ...
7 AFNI output
8 ...
9 ...
10 [user@compute200 ~]$ exit
11 [user@submitt0 ~]$
```

✖ Srun will automatically close your session after 9 hours

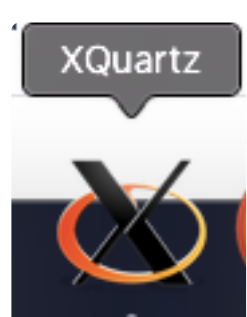
macOS

Pre-Requisites

- Install XQuartz through the [Workspace One Intelligent Hub](#)
- [Configure the terminal](#) for X11 Forwarding

Instructions

1. Start XQuartz if you don't already have it running



2. SSH to submit0 with the -Y argument to open an SSH session with X11 forwarding

```
1 ssh -YC submit0.laureateinstitute.org
```

At this point you are connected to the Submit node and are running commands on the analysis submit node.

3. You can verify that your X11 forwarding is enabled by printing the DISPLAY variable (this will be empty if it isn't enabled)

```
1 [user@submit0 ~]$ echo $DISPLAY
2 10.124.152.6:10.0
```

4. Use `srun` to open an interactive bash shell with X11 forwarding on a compute node

```
1 srun --pty --x11 bash
2 # --pty opens an interact-able bash shell and --x11 enables X11 forwarding
```

5. From here you can load the AFNI module and then run the afni command to open the GUI, or run any other program you would like.

```
1 module load afni
2 afni
```

6. Close the AFNI window when you are done and enter the exit command into the command line when you are finished.

Example

Example output

```
1 [user@subm10 ~]$ srun --pty --x11 bash
2 srun: error: x11: job has no allocated nodes defined
3 [user@compute200 ~]$ module load afni
4 [user@compute200 ~]$ afni
5 ...
6 ...
7 AFNI output
8 ...
9 ...
10 [user@compute200 ~]$ exit
11 [user@subm10 ~]$
```

✖ Srun will automatically close your session after 9 hours

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